

Problem

Consider the boundary-value problem shown in the textbook:

$$\frac{\partial^2 u_4}{\partial x^2} + \frac{\partial^2 u_4}{\partial y^2} = 0$$

on the rectangular domain

$$0 \leq x \leq L, \quad 0 \leq y \leq H,$$

with boundary conditions

$$u_4(0, y) = g_1(y), \tag{BC1}$$

$$u_4(L, y) = 0, \tag{BC2}$$

$$u_4(x, 0) = 0, \tag{BC3}$$

$$u_4(x, H) = 0. \tag{BC4}$$

The goal is to derive the solution using *exponential functions*, rather than writing the intermediate solution in terms of hyperbolic functions.

1 Separation of Variables

Assume a product solution

$$u_4(x, y) = h(x)\phi(y).$$

Then

$$\frac{h''(x)}{h(x)} = -\frac{\phi''(y)}{\phi(y)}.$$

Thus the two ordinary differential equations are

$$\boxed{h'' - \lambda h = 0} \tag{1}$$

and

$$\boxed{\phi'' + \lambda \phi = 0.} \tag{2}$$

2 Boundary Conditions for $\phi(y)$

From BC3,

$$u_4(x, 0) = h(x)\phi(0) = 0.$$

For a nontrivial solution,

$$\boxed{\phi(0) = 0}.$$

Similarly, from BC4,

$$u_4(x, H) = h(x)\phi(H) = 0,$$

so

$$\boxed{\phi(H) = 0}.$$

Therefore,

$$\phi'' + \lambda \phi = 0, \quad \phi(0) = \phi(H) = 0.$$

Eigenvalues and eigenfunctions:

$$\lambda_n = \left(\frac{n\pi}{H}\right)^2,$$

$$\phi_n(y) = \sin \frac{n\pi y}{H}, \quad n = 1, 2, \dots$$

3 Solve $h_n(x)$ Using Exponentials

For the n -th eigenvalue,

$$\lambda_n = \left(\frac{n\pi}{H}\right)^2.$$

The h -equation

$$h'' - \lambda h = 0$$

therefore becomes

$$h_n'' - \left(\frac{n\pi}{H}\right)^2 h_n = 0.$$

Define

$$k_n = \frac{n\pi}{H}.$$

Then

$$h_n'' - k_n^2 h_n = 0.$$

The characteristic equation is

$$r^2 - k_n^2 = 0,$$

so

$$r = \pm k_n.$$

Thus the exponential general solution is

$$h_n(x) = A_n e^{k_n x} + B_n e^{-k_n x}. \quad (3)$$

4 Apply $h_n(L) = 0$

From BC2,

$$h_n(L) = 0.$$

Therefore,

$$A_n e^{k_n L} + B_n e^{-k_n L} = 0.$$

Solving for B_n ,

$$B_n e^{-k_n L} = -A_n e^{k_n L},$$

and hence

$$B_n = -A_n e^{2k_n L}.$$

Thus

$$h_n(x) = A_n e^{k_n x} - A_n e^{2k_n L} e^{-k_n x},$$

or

$$h_n(x) = A_n \left(e^{k_n x} - e^{k_n(2L-x)} \right). \quad (4)$$

5 Choose the Exponential Form that Matches the Textbook

At this point,

$$h_n'' - k_n^2 h_n = 0, \quad k_n = \frac{n\pi}{H},$$

with

$$h_n(L) = 0.$$

Starting from the exponential general solution,

$$h_n(x) = A_n e^{k_n x} + B_n e^{-k_n x},$$

the condition $h_n(L) = 0$ gives

$$A_n e^{k_n L} + B_n e^{-k_n L} = 0,$$

and therefore

$$B_n = -A_n e^{2k_n L}.$$

It is convenient to rewrite the resulting solution in terms of $x - L$. **Since the multiplicative constant is arbitrary**, we may write

$$h_n(x) = A_n \frac{e^{k_n(x-L)} - e^{-k_n(x-L)}}{2}. \quad (5)$$

This expression satisfies

$$h_n(L) = A_n \frac{1 - 1}{2} = 0$$

automatically.

Substituting $k_n = n\pi/H$,

$$h_n(x) = A_n \frac{e^{\frac{n\pi}{H}(x-L)} - e^{-\frac{n\pi}{H}(x-L)}}{2}. \quad (6)$$

The factor 1/2 is chosen deliberately so that, after applying $h_n(0)$, the coefficient A_n will be exactly the same coefficient used in the textbook's sinh-form.

6 Construct the Product Solutions

For each n ,

$$u_{4,n}(x, y) = h_n(x) \phi_n(y).$$

Using (6) and

$$\phi_n(y) = \sin \frac{n\pi y}{H},$$

we obtain

$$u_{4,n}(x, y) = A_n \sin \frac{n\pi y}{H} \frac{e^{\frac{n\pi}{H}(x-L)} - e^{-\frac{n\pi}{H}(x-L)}}{2}. \quad (7)$$

Because Laplace's equation is linear, we superpose these product solutions:

$$u_4(x, y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi y}{H} \frac{e^{\frac{n\pi}{H}(x-L)} - e^{-\frac{n\pi}{H}(x-L)}}{2}. \quad (8)$$

7 Determine the Fourier Coefficients

Apply BC1:

$$u_4(0, y) = g_1(y).$$

Setting $x = 0$ in (8),

$$g_1(y) = \sum_{n=1}^{\infty} A_n \frac{e^{-\frac{n\pi L}{H}} - e^{\frac{n\pi L}{H}}}{2} \sin \frac{n\pi y}{H}. \quad (9)$$

Multiply both sides by

$$\sin \frac{m\pi y}{H}$$

and integrate from 0 to H :

$$\int_0^H g_1(y) \sin \frac{m\pi y}{H} dy = \sum_{n=1}^{\infty} A_n \frac{e^{-\frac{n\pi L}{H}} - e^{\frac{n\pi L}{H}}}{2} \int_0^H \sin \frac{n\pi y}{H} \sin \frac{m\pi y}{H} dy.$$

Using the orthogonality relation

$$\int_0^H \sin \frac{n\pi y}{H} \sin \frac{m\pi y}{H} dy = \begin{cases} 0, & n \neq m, \\ \frac{H}{2}, & n = m, \end{cases}$$

only the $n = m$ term remains:

$$\int_0^H g_1(y) \sin \frac{m\pi y}{H} dy = A_m \frac{e^{-\frac{m\pi L}{H}} - e^{\frac{m\pi L}{H}}}{2} \frac{H}{2}.$$

Therefore,

$$A_m = \frac{4}{H \left(e^{-\frac{m\pi L}{H}} - e^{\frac{m\pi L}{H}} \right)} \int_0^H g_1(y) \sin \frac{m\pi y}{H} dy. \quad (10)$$

Since m is an arbitrary positive integer, we may write

$$A_n = \frac{4}{H \left(e^{-\frac{n\pi L}{H}} - e^{\frac{n\pi L}{H}} \right)} \int_0^H g_1(s) \sin \frac{n\pi s}{H} ds. \quad (11)$$

Here s is a dummy integration variable.

8 Final Solution in Exponential Form

Substituting (11) into (8), the complete solution is

$$u_4(x, y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi y}{H} \frac{e^{\frac{n\pi}{H}(x-L)} - e^{-\frac{n\pi}{H}(x-L)}}{2}, \quad (12)$$

where

$$A_n = \frac{4}{H \left(e^{-\frac{n\pi L}{H}} - e^{\frac{n\pi L}{H}} \right)} \int_0^H g_1(s) \sin \frac{n\pi s}{H} ds. \quad (13)$$

9 Recovering Exactly the Textbook's Solution

Recall

$$\sinh z = \frac{e^z - e^{-z}}{2}.$$

Thus

$$\frac{e^{\frac{n\pi}{H}(x-L)} - e^{-\frac{n\pi}{H}(x-L)}}{2} = \sinh\left(\frac{n\pi}{H}(x-L)\right).$$

Therefore (12) becomes exactly

$$u_4(x, y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi y}{H} \sinh\left(\frac{n\pi}{H}(x-L)\right),$$

and

$$A_n = \frac{2}{H \sinh\left(-\frac{n\pi L}{H}\right)} \int_0^H g_1(y) \sin \frac{n\pi y}{H} dy. \quad (16)$$

Final Comparison

The textbook solution is

$$u_4(x, y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi y}{H} \sinh\left(\frac{n\pi}{H}(x-L)\right),$$

with

$$A_n = \frac{2}{H \sinh\left(\frac{n\pi(-L)}{H}\right)} \int_0^H g_1(y) \sin \frac{n\pi y}{H} dy.$$

The completely exponential version is

$$u_4(x, y) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi y}{H} \frac{e^{\frac{n\pi}{H}(x-L)} - e^{-\frac{n\pi}{H}(x-L)}}{2},$$

with

$$A_n = \frac{4}{H \left(e^{-\frac{n\pi L}{H}} - e^{\frac{n\pi L}{H}} \right)} \int_0^H g_1(y) \sin \frac{n\pi y}{H} dy.$$

These two formulas are exactly the same because

$$\sinh z = \frac{e^z - e^{-z}}{2}.$$